Narges Sefid, MSc

Energy Systems Modeler

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I build capacity-expansion and long-term planning models in Python, and use them to study how energy systems can become truly sustainable – which technologies, where, at what cost, and under which land-use and policy constraints.

Experience

Energy Modelling Intern Sep 2022 - Aug 2024

Net Zero Atlantic, Halifax, NS/Hybrid

- Integrated **land-use** constraints into the **TEMOA-ACES** energy model in **Python** to evaluate Net Zero trade-offs across four **Atlantic provinces** and 17 energy production technologies, and studied 20+ potential land use scenarios by 2050 for **long-term resource planning**. (Presented at the **Energy Modeling Hub conference**, Moncton, June 2024.) (Code)
- Developed a **GIS-based spatial analysis** in **Python** which combines over 5 datasets and 8 land-use layers, analysing 600+ viable renewable sites in terms of their capacity and **transmission cost** to inform long-term planning and cost-optimization.
- Processed **ERA5** hourly reanalysis data in **Python**, **clustering** them into seven representative days to support future energy system development planning and forecasting.
- Contributed to a provincial **proposal** supporting **climate policy** in **Prince Edward Island**.

Research and Teaching Assistant Jan 2022 - Dec 2024

Simon Fraser University, Surrey, BC

- Conducted **labs and tutorials** on the **OSeMOSYS** energy model and supported courses on **sustainable product design** and **sustainable business development**.
- Community and **Stakeholder Engagement** for **Equity** and **Justice** in Energy Modelling. (Manuscript)

Researcher and Project Manager Jul 2017 - Oct 2021

Manasazan Sadid, Tehran, Iran

- Led the project for Iran's **national standard on energy consumption labels** for coolers. Chaired a committee of 20+ members to design a **test plan** and implemented the plan in collaboration with 10+ vendors on 20+ test samples.

Energy Market Analyst Dec 2017 - Sep 2018

Ariosunshade, Tehran, Iran

- Analyzed the energy market for various solar energy technologies.
- Commercialized a state-of-the-art thin-film photovoltaic panel.

Education

MSc, Sustainable Energy Engineering 2022 - 2026

Simon Fraser University, Surrey, BC

Energy systems modelling, long-term planning, and resource-development analysis.

Key courses: Geographical Information Science (GIS), Machine Learning for Sustainable Systems, Modelling for SDGs

MSc, Energy Systems Engineering 2015 - 2018

University of Tehran, Tehran, Iran

Key courses: Energy Systems Analysis, Programming for Advanced Mathematics

Selected Projects

- **Land Limits in TEMOA-ACES** — Land-use constraints and 20+ scenarios for Atlantic Canada's net-zero pathways, in Python. (Code)

- **Wind & Agriculture Co-location, PEI** — Onshore wind and agriculture co-location scenarios under high renewable penetration – my MSc thesis work, presented at the Energy Modeling Hub conference (Ottawa, 2025).
- **Community Engagement in Energy Modelling** — Research on community and stakeholder engagement for equity and justice in energy modelling, and what it changes in practice. (Manuscript)
- **Wind Speed Forecasting with ML** — Machine-learning forecasting of long-term wind speed under climate change scenarios, in Python. (Code)
- **Steel Industry Decarbonization Model** — A national techno-economic energy system model minimizing CO_2 emissions from iron and steel, through scenario analysis. (Code)

Skills

Energy Modelling: TEMOA · OSeMOSYS · CLEWs · capacity expansion · cost-optimization / production-cost analysis · scenario analysis
Programming: Python (primary – all my modelling) · GAMS (Beginner) · MATLAB · R (Beginner) · Git/GitHub
Data & Spatial Analysis: GIS · clustering · time series · forecasting · SQL · machine learning · Excel
Communication: Technical writing · presenting to technical, stakeholder & policy audiences · teaching · LaTeX

Certifications

Sustainability in Energy Micro-Credential, 2025.

Publications & Presentations

- **Manuscript** – Community and Stakeholder Engagement for Equity and Justice in Energy Modelling. (SFU Summit)
- **Poster** – Onshore wind energy and land impacts in the Atlantic Canada Energy System (TEMOA-ACES). Energy Modeling Hub Conference, Moncton, 2024.
- **Poster & pitch** – Spatial and land-use considerations of high renewable penetration (Prince Edward Island case). Energy Modeling Hub Annual Conference, Ottawa, 2025.

Teaching

- **Integrated Energy Systems** (SFU, SEE, May 2023) – labs and tutorials on the OSeMOSYS energy model.
- **Integrated Energy Solution 1** (SFU, SEE) – sustainable product design: sessions, materials, and student project support.
- **The Business of Engineering** (SFU, SEE) – sustainable business development and the economics of engineering.

Professional Engagement

- **Student Delegate** – COP30, with Simon Fraser University.
- **Volunteer OSeMOSYS Troubleshooting Trainer** – EMPLAC 2022, Climate Compatible Growth, Online – helping modellers debug and troubleshoot their OSeMOSYS energy models.
- **Communication Lead, Delta E Plus Research Group** – Managed and improved team communication channels, including social media platforms, enhancing the clarity, reach, and accessibility of research-related content.
- **Guest Mentor** – Opportunity Fest Innovation Competition, Simon Fraser University, BC.
- **Visiting Researcher** – NTNU, Trondheim, Norway (Nov–Dec 2025). Knowledge-exchange visit – presenting my research and joining presentations and discussions on NTNU's research.